

Chenzhe Zhang

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Aug 2025-Dec 2026

Master of Science in Civil and Computer Engineering

University of Illinois Urbana-Champaign, Champaign, IL

Sept 2021-Jun 2025

Bachelor of Science in Civil Engineering

- Major: Structural Engineering Minor: Transportation Engineering

EXPERIENCE

Zhejiang Zhongcheng Construction Group

Special-Grade General Contractor; affiliated with Zhongcheng Holding Group, a China Top 500 Enterprise

Exposure across bidding-stage BIM and on-site construction for the same large-scale project

Project Department Intern

May 2026 – Present

- Compared construction drawings with field conditions to understand structural components, construction sequences, and on-site execution methods during the construction phase.
- Reviewed discrepancies between design documents and field implementation to better understand drawing interpretation, construction coordination, and practical constraints during project delivery.
- Participated in project management support tasks, including tracking construction progress, reviewing site organization, assisting with construction documentation, and observing coordination among technical, engineering, and project teams.

BIM Technology Center Intern

Jun 2025 – Jul 2025

- Supported bidding-stage BIM visualization by translating complex CAD drawings into Revit-based architectural, structural, facade, foundation pit support, and site logistics models.
- Developed schematic BIM models for building massing, aluminum panel facade systems, roof structures, structural beams and columns, and steel-frame–reinforced concrete core-tube components.
- Used Revit and Lumion to produce 3D construction-condition visualizations, helping communicate excavation sequence, site organization, equipment circulation, and key construction stages.
- Assisted in preparing technical documentation for a foundation pit support concept involving segmented water-stop inclined piles and vertical retaining systems, with emphasis on construction feasibility and engineering application scenarios.

Zhejiang Provincial Institute of Land Spatial Planning Intern

Jun 2023–Aug 2023

- Used ArcGIS to analyze spatial, transportation, and land-use data, supporting regional infrastructure planning and transportation route optimization.
- Conducted suitability and accessibility analyses and produced 10+ thematic maps to support corridor siting, route adjustments, and planning documentation.

PROJECTS

Optimization Design of Order Dispatch System

Dec 2024–Jun 2025

- Developed a Python-based simulation and multi-objective routing optimization model to evaluate delivery distance, punctuality, cost, and workload fairness.

Integration of 3D Printed Fiber Concrete and OpenSees to Enhance Structure Design

Jun 2024–Dec 2024

- Designed 3D-printed fiber-reinforced concrete specimens, collected strain data, derived a constitutive model, and implemented it in OpenSees to support structural analysis and topology optimization.

COMPUTER SKILLS & OTHERS

BIM & Design Tools | Revit, AutoCAD, Lumion, ArcGIS, OpenSees

Programming & Analysis | Python, SQL, PyTorch, PyQt5, MATLAB

Construction & Civil Skills | BIM Modeling, CAD Drawing Interpretation, Construction Documentation, Site Logistics, Foundation Pit Support, Structural Analysis

Other Tools | MS Office, Photoshop, Adobe Illustrator